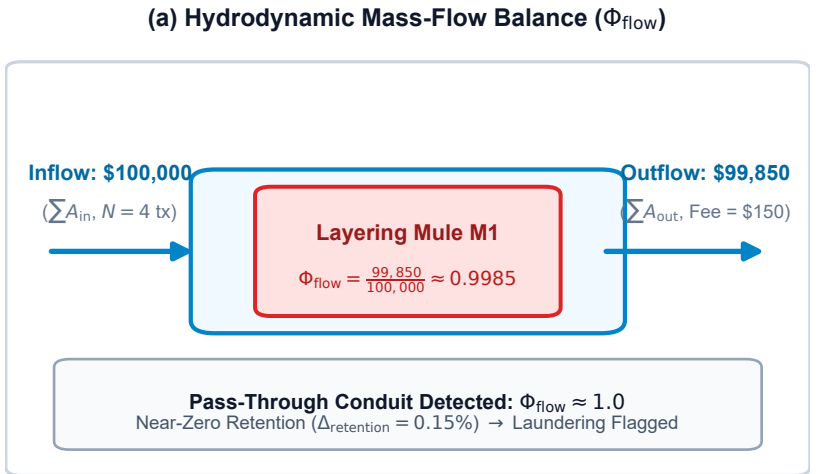
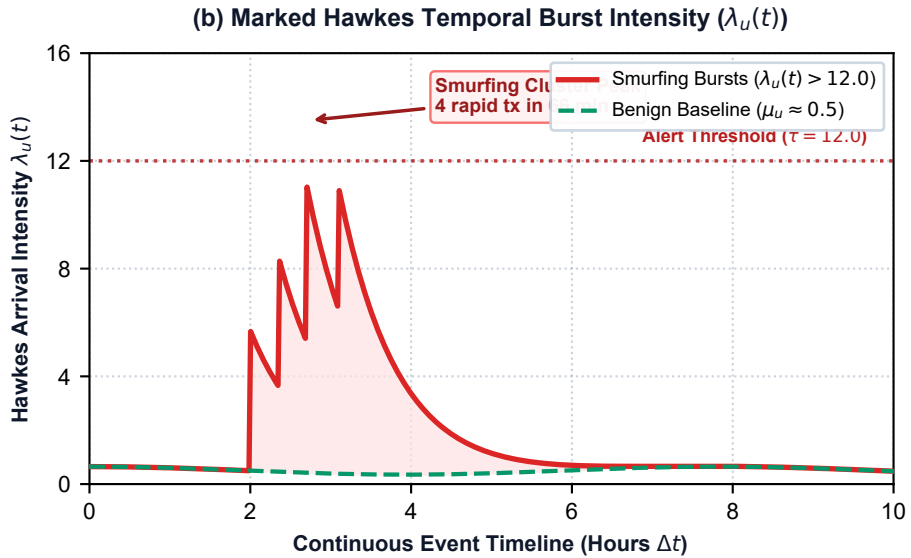
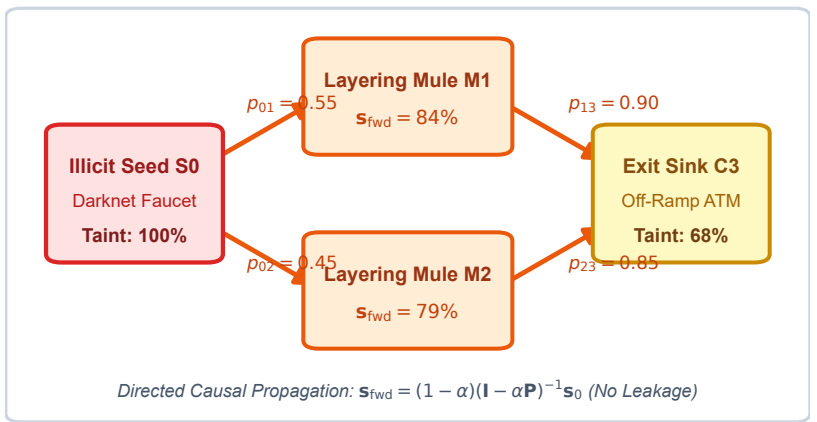
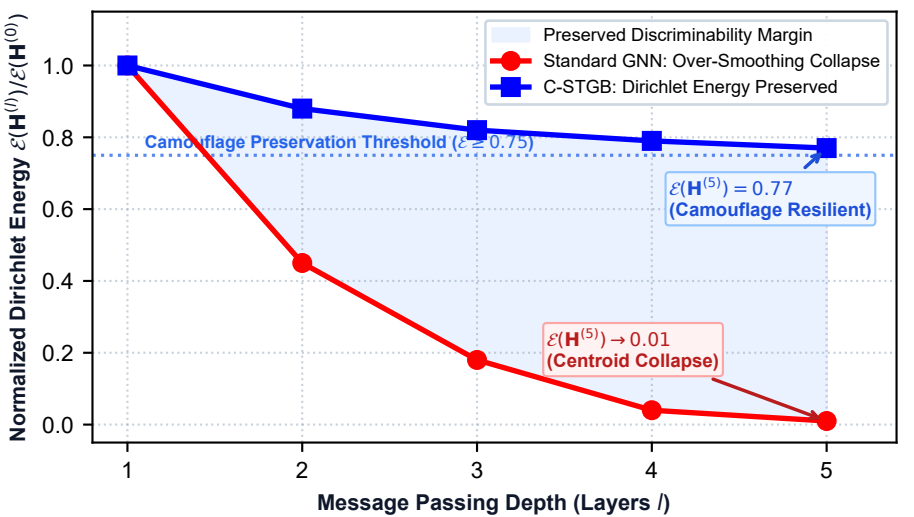

(a) Hydrodynamic Mass-Flow Balance (Φ_{flow})
Inflow: $100,000
($\sum A_{in}$, $N = 4$ tx)
Layering Mule M1
$\Phi_{flow} = \frac{99,850}{100,000} \approx 0.9985$
Outflow: $99,850
($\sum A_{out}$, Fee = $150)
Pass-Through Conduit Detected: $\Phi_{flow} \approx 1.0$
Near-Zero Retention ($\Delta_{retention} = 0.15\%$) → Laundering Flagged


(b) Marked Hawkes Temporal Burst Intensity ($\lambda_u(t)$)
Smurfing Bursts ($\lambda_u(t) > 12.0$)
Benign Baseline ($\mu_u \approx 0.5$)
Alert Threshold ($\tau = 12.0$)
Hawkes Arrival Intensity $\lambda_u(t)$
Continuous Event Timeline (Hours Δt)


(c) Directed Causal Taint Exposure ($\mathbf{s}_{fwd}, \mathbf{s}_{rev}$)
Illicit Seed S0
Darknet Faucet
Taint: 100%
Layering Mule M1
$\mathbf{s}_{fwd} = 84\%$
Layering Mule M2
$\mathbf{s}_{fwd} = 79\%$
Exit Sink C3
Off-Ramp ATM
Taint: 68%
$p_{01} = 0.55$
$p_{02} = 0.45$
$p_{13} = 0.90$
$p_{23} = 0.85$
Directed Causal Propagation: $\mathbf{s}_{fwd} = (1-\alpha)(\mathbf{I} - \alpha\mathbf{P})^{-1}\mathbf{s}_0$ (No Leakage)


(d) Spectral Dirichlet Energy Preservation ($\mathcal{E}(\mathbf{H})$)
Preserved Discriminability Margin
Standard GNN: Over-Smoothing Collapse
C-STGB: Dirichlet Energy Preserved
Camouflage Preservation Threshold ($\varepsilon \geq 0.75$)
$\mathcal{E}(\mathbf{H}^{(5)}) = 0.77$
(Camouflage Resilient)
$\mathcal{E}(\mathbf{H}^{(5)}) \to 0.01$
(Centroid Collapse)
Normalized Dirichlet Energy $\mathcal{E}(\mathbf{H}^{(l)})/\mathcal{E}(\mathbf{H}^{(0)})$
Message Passing Depth (Layers l)